土木工程学院+混凝土基本原理上+《混凝土及砌体结构上第2版》三四五七章课后答案

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Torch Note Club

第3章

(1)解 $g_k = 25\times0.07\times0.6 + 20\times0.025\times0.6 + 16\times0.015\times0.6$

恒荷载 $= 1.494\ kN/m$

$q_k = 2\times0.6 = 1.2\ kN/m$

(2)解 $M_k = \frac{1}{8}(g_k+q_k)l_0^2 = \sim\sim = 2.64\ kN\cdot m$

$M = \gamma_0 \frac{1}{8}(1.3g_k+1.5q_k)l_0^2 = \sim = 3.67\ kN\cdot m$

γ_0 ↓ 查表3-5，本题已给出

第4章

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4-1 解：图参考93页

$a_s = 40\ mm$ ↓ 查表，在范围内就行　　$h_0 = h - a_s = \cdots = 560mm$

$$\alpha_s = \frac{M}{\alpha_1 f_c b h_0^2} = \cdots = 0.152$$

$$\xi = 1-\sqrt{1-2\alpha_s} = \cdots = 0.165 < \xi_b = 0.518$$

↓ 查表，未超筋

$$A_s = \frac{\alpha_1 f_c b h_0 \xi}{f_y} = \cdots = 919\ mm^2$$

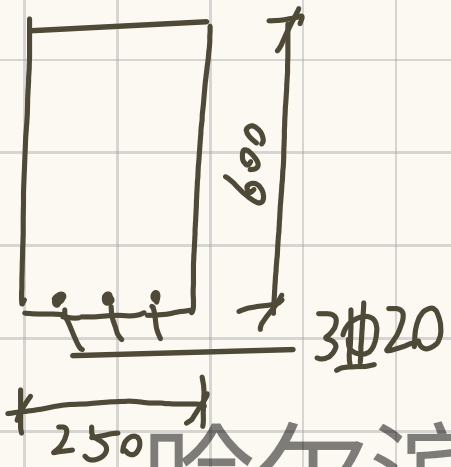
$A_s > \rho_{min} bh = \min(0.2\%, 0.45\frac{f_t}{f_y}) = \cdots 300mm^2$

未少筋

3⌀20 ($A_s = 942 mm^2$)

也可 γ_s, $A_s = \frac{M}{f_y \gamma_s h_0}$

考试不给



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4-2

$a_s = 20mm$ $h_0 = 200 - a_s = 80 mm$

$\alpha_s = \cdots = \cdots = 0.197$

$\xi = \cdots = \cdots = 0.222 < \xi_b = 0.518$ 未超筋

$A_s = \cdots = \cdots = 587 mm^2$

$A_s > \rho_{min} bh = 200 mm^2$

9⌀10 ($A_s = 604 mm^2$

画图

4-3

$A_s = 804\,mm^2 > \rho_{min} bh = \cdots = 180\,mm^2$

$h_0 = 450 - 40 = 410\,mm$

$\xi = \dfrac{f_y A_s}{\alpha_1 f_c b h_0} = \cdots = 0.211 < \xi_b = 0.518$

$M_u = \xi(1-0.5\xi)\alpha_1 f_c b h_0^2 = \sim = 106.13\,kN\cdot m$

$> M = 105\,kN\cdot m$

$M = \gamma_0 \frac{1}{8}(1.3g_k + 1.5q_k) l_0^2$

4-4

$a_s = 65\,mm \qquad h_0 = \sim = 435\,mm$

$M_{max} = \alpha_1 f_c b h_0^2 \xi_b (1-0.5\xi_b) = \cdots = 228.5\,kN\cdot m$

$A_s' = \dfrac{M - M_{max}}{f_y'(h_0 - a_s)} = \cdots = 363\,mm^2$

$A_s = \dfrac{\alpha_1 f_c b h_0 \xi_b + f_y' A_s'}{f_y} = \cdots = 2331\,mm^2$

2⌀16，5⌀25

4-5

$M_1 = f_y' A_s' (h_0 - a_s') = 72.38\,kN\cdot m$

$M_2 = M - M_1 = \cdots = 207.62\,kN\cdot m$

$\alpha = \dfrac{M_2}{\cdots} = \cdots = 0.349$

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$$\alpha_{s2} = \frac{\quad}{\alpha_1 f_c b h_0^2} = \cdots = 0.511$$

$$\xi = \cdots = \cdots = 0.420 < \xi_b = 0.518$$

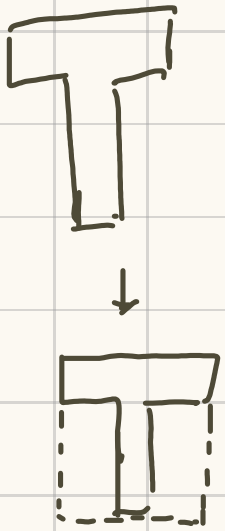
$$x = \xi h_0 = \cdots = 196\,mm > 2a_s' = 80\,mm$$

$$A_s = \frac{\alpha_1 f_c b x + f_y A_s'}{f_y} = \cdots = 2220\,mm^2$$

6 Φ22

4-6

$a_s = 40mm$ $\quad h_0 = 410mm$



$$\alpha_1 f_c b_f' h_f' \left(h_0 - \frac{h_f'}{2}\right) = \cdots = 876.75\,kN\cdot m$$

$> M = 90\,kN\cdot m$

一类截面梁

$$\alpha_s = \frac{M}{\alpha_1 f_c b_f' h_0^2} = \cdots = 0.016$$

$$\xi = 1 - \sqrt{1 - 2\alpha_s} = \cdots = 0.016 < \xi_b$$

$$A_s = \cdots = \cdots = 615\,mm^2$$

$$A_s > \rho_{min} bh$$

4 Φ14

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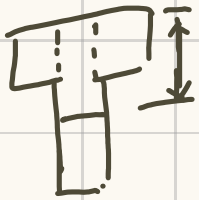
4-7 $a_s = 65mm$ $h_0 = 435mm$

$\alpha_1 f_c b_f' h_f' (h_0 - \frac{1}{2}h_f') = \cdots = 220.22 kN\cdot m$

$< M = 300 kN\cdot m$



二类

$$\alpha_s = \frac{M - \alpha_1 f_c (b_f' - b) h_f' (h_0 - \frac{h_f'}{2})}{\alpha_1 f_c b h_0^2}$$

$= 0.351$

$\xi = 1 - \sqrt{1 - 2\alpha_s} = \cdots = 0.454 < \xi_b = 0.518$

$$A_s = \frac{\alpha_1 f_c b h_0 \xi + \alpha_1 f_c (b_f' - b) h_f'}{f_y} = \cdots = 2363 \ m^2$$

5⌀25

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如何判断几排钢筋

1、看力的大小：力大两排，>4根配两排

2、按1排，结果 A_s ↑5%

不行，四

第 5 章

5-1 $h_0 = 450 - 40 = 410\,mm$

验算截面 $V = 88kN < 0.25\beta_c f_c b h_0 = \cdots = 263.8kN$

$0.7 f_t b h_0 = \cdots = 73.87kN < V = 88kN$

$$\frac{A_{sv}}{s} = \frac{V - 0.7 f_c b h_0}{f_{yv} h_0} = 0.128$$

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选用双肢 $\phi 8$

$$s = \frac{A_{sv}}{0.128} = \frac{2 \times 50.3}{0.128} = 786\,mm$$

取 $s = s_{max} = 200\,mm$

$$\rho_{sv} = \frac{n A_{sv1}}{bs} = \frac{2 \times 50.3}{180 \times 200} = 0.28 >$$

$$\rho_{sv,min} = 0.24 \frac{f_t}{f_{yv}} = \cdots = 0.13\%$$

$\phi 8 @ 200$

5-2 $V = \frac{1}{2} q l_n = \frac{1}{2} \times 60 \times 5.5 = 165 kN$

剪力用 5.5
弯矩用 5.74

$a_s = 40$, $h_0 = \cdots = 510 mm$

$\frac{h_w}{b} = \cdots = 2.04 < 4$,

$V < 0.25 \beta_c f_c b h_0 = \cdots = 455.81 kN$

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$0.7 f_t b h_0 = \cdots = 127.6 kN < V$

$\Phi 6 @ 180$

$\rho_{sv} = \frac{A_{sv}}{bs} = 0.16\% > \rho_{sv,min} = \cdots = 0.11\%$

$V_{cs} = 0.7 f_t b h_0 + \frac{f_{yv} A_{sv}}{s} h_0 = \cdots = 175.71 kN$
$> 165 kN$

故不需配弯起钢筋

易错点：ϕ Φ Φ

5-3 四、五章结合，弯起钢筋为纵向受拉…

(1) $V = 110 + \frac{1}{2} \times 10 \times 4.5 = 132.5\ kN$

$$\frac{110}{132.5} = 0.83 > 0.75$$

$$\frac{h_w}{b} = \cdots = 2.04 < 4$$

$$V < 0.25\beta_c f_c b h_0 = 455.81\ kN$$

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$$\lambda = \frac{a}{h_0} = \frac{1500}{510} = 2.94$$

$$\frac{1.75}{\lambda+1} f_t b h_0 = \cdots = 80.98\ kN < V$$

$$\frac{A_{sv}}{s} = \frac{V - \frac{1.75}{\lambda+1} f_t b h_0}{f_{yv} h_0} = 0.337$$

取 双肢 Φ 6

$$s = \frac{A_{sv}}{0.337} = \cdots = 168\ mm$$

取 $s = 160\ mm$ （s应往小取）

s取整数

验算 $\rho_{sv} = \cdots > \rho_{sv,min}$

双肢 Φ6 @160

(2) $M = \frac{1}{8}ql^2 + P \cdot a = \cdots = 190.3\ kN$

$h_0 = 510\ mm$

$\alpha_s = \frac{M}{\alpha_1 f_c b h_0^2} = \cdots = 0.205$

$\xi = 1 - \sqrt{1-2\alpha_s} = \cdots = 0.232 < \xi_b$

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$A_s = \frac{\alpha_1 f_c b h_0 \xi}{f_y} = \cdots = 1115\ mm^2$

ρ_{min} 验算 $\Rightarrow 4\Phi 20$

$\rho_{sv} = \frac{2A_{sv1}}{bs} = \cdots 0.12\% > \rho_{sv,min} = \cdots = 0.114\%$

配箍率仍需验算

$V = \frac{1.75}{\lambda+1} f_t b h_0 + \frac{f_{yv} A_{sv}}{s} h_0 = \cdots = 126.6\ kN$

$cs \quad \lambda+1 \quad c \qquad s$

$$A_{sb} = \frac{V - V_{cs}}{0.8 f_y \sin\alpha} = \cdots = 29\,mm^2$$

弯起一根钢筋即可

(然后计算是否需要弯起第二根钢筋)

$$V_2 = 132.5 - 15 \times \frac{690}{1500} = 125.6\,kN < V_{cs} = 126.6$$

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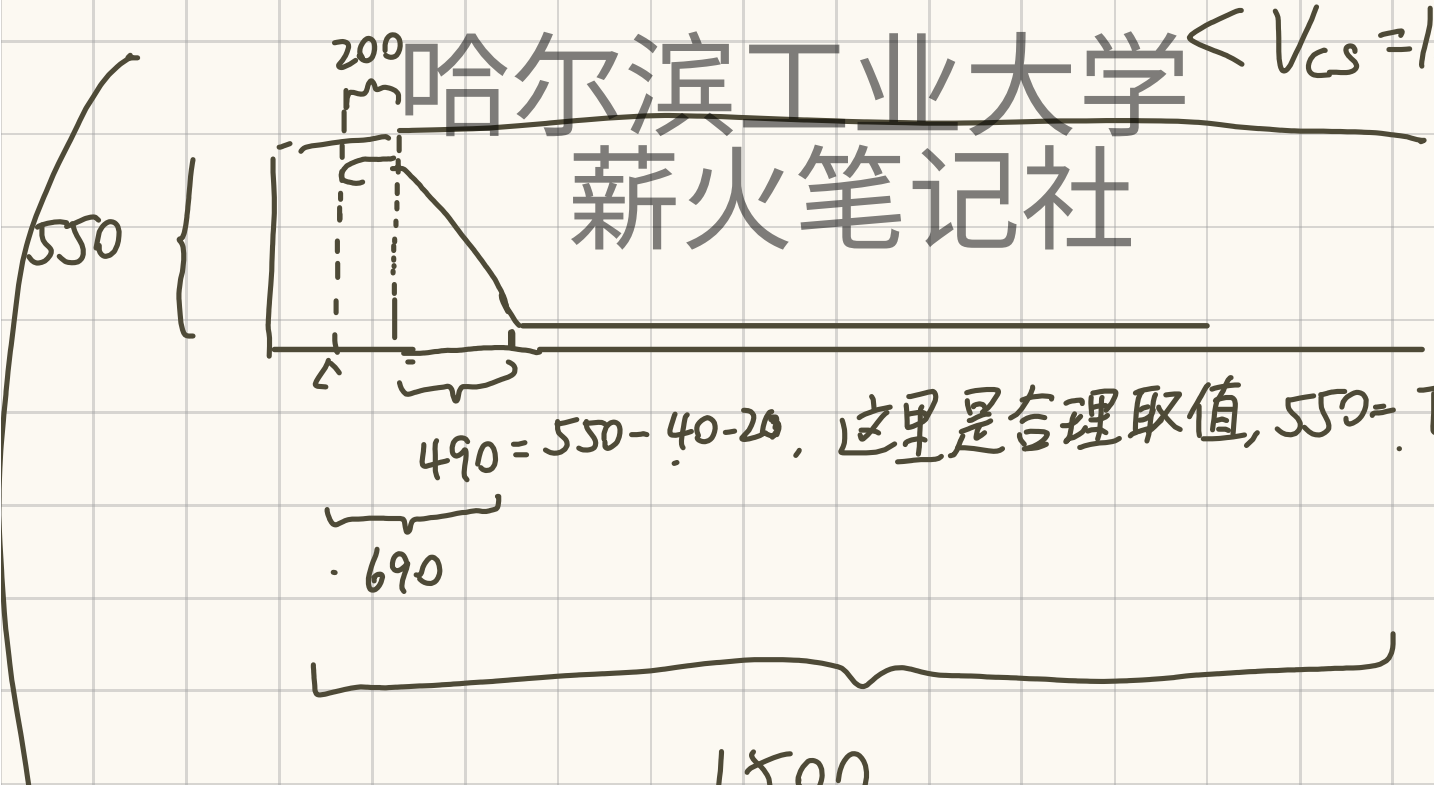
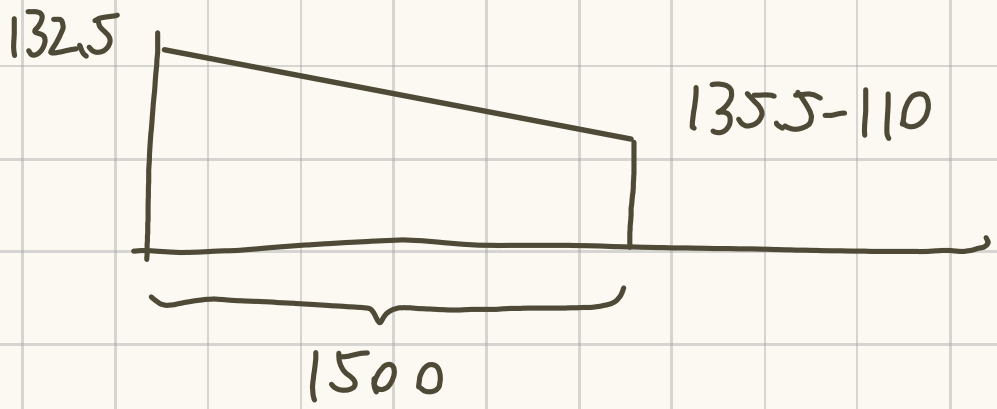


剪力图



$690 = 490 + 200$ 见图

$15 = q \cdot l' = 10 \times 1.5m = 15$

5-4

(1) $a_s = 40$, $h_0 = \cdots = 460\,mm$

$V_u = 0.7 f_t b h_0 + f_{yv} \frac{A_{sv}}{s} h_0 = \cdots = 203.15\,kN$

验算 $0.25\beta_c f_c b h_0$, $\rho_{sv,min}$

(2) $\frac{1}{2} q l_0 = 203.15\,kN$

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$q = 84.65\,kN/m$

混凝土自重 $25\,kN/m^3$

$25 \times 0.2 \times 0.5$

注意：$q = \gamma_G g_k + \gamma_Q g_k$

$\gamma_G \to 1.3$, $\gamma_Q \to 1.5$

$q_k = 54.27\,kN/m$

第7章

7-1 解 $\frac{l_0}{b} = \frac{6400}{400} = 16 \Rightarrow \varphi = 0.87$

矩形截面

$$A_s' = \frac{\frac{N}{0.9\varphi} - f_c A}{f_y'} = \cdots = 915\ mm^2$$

$$< 3\% \times 400 \times 400 = 4800\ mm$$

$$A_{s\,min}' = 0.55\% \times 400 \times 400 = 880 < 915\ mm^2$$

$4\Phi 18$. $A_s' = 1018\ mm^2$

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2、 $l_0/d = \cdots = 12$ $\rho' = 3\% \Rightarrow A_s' = 0.03A = 5887.5$

假定的

选 $12\Phi 25$

$$d_{cor} = 500 - 2\times(20 + \tfrac{1}{2}) = 436\ mm^2$$

保护层 钢筋直径一半

$$A_{cor} = \frac{\pi d_{cor}^2}{4} = \cdots = 149225\ mm^2$$

$$A_{sso} = \frac{\frac{N}{0.9} - (f_c A_{cor} + f_y A_s)}{2\alpha f_{yv}} = \cdots = 2356\ mm^2$$

$$> 0.25 A_s' = 1473\ mm^2$$

$$\Rightarrow \Phi 12 \Rightarrow A_{ss1} = 113\ mm^2 \Rightarrow s = \frac{\pi d_{cor} A_{ss1}}{A_{sso}} = 66\ mm$$

取 $s = 60\ mm$

$$A_{sso} = \frac{\pi d_{cor} A_{ss1}}{s} = 2578\ mm^2$$

$$N_u^{螺} = 0.9(f_c A_{cor} + f_y' A_s' + 2\alpha f_{yv} A_{sso}) = \cdots = 6144.1\ kN$$

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$$N_u^{普} = 0.9\varphi(f_c A + f_y A_s') = \cdots = 4766.2\ kN \times 1.5$$

$$> N_u^{螺}$$

$$N_u = N_u^{螺} = 6144.1\ kN > 6000\ kN = N$$

7.3 $a_s = 40\ mm$

$$h_0 = h - a_s = 460\ mm$$

$$\frac{M_1}{M_2} = \cdots = 0.67 < 0.9$$

$$\frac{N}{f_c bh} = \cdots = 0.699 < 0.7$$

$$\frac{l}{h} = \frac{4500}{500} = 9 > 9.8 - 3.5\frac{M_1}{M_2} = 7.47$$

考虑二阶效应

$$C_m = 0.7 + 0.3\frac{M_1}{M_2} = \cdots = 0.9$$

$$\zeta = \frac{0.5 f_c A}{N} = \cdots = 0.72$$

$$\eta_{ns} = 1 + \frac{1}{\frac{1300\left(\frac{M_2}{N} + e_a\right)}{h_0}} \cdot \left(\frac{l_c}{h}\right)^2 \zeta = \cdots = 1.09$$

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$$e_a = \max\left\{\frac{h}{30}, 20\right\} = 20\,mm$$

$$C_m \eta_{ns} = \cdots = 0.98 < 1 \Rightarrow C_m \eta_{ns} = 1.0$$

$$M_1 = C_m \eta_{ns} M_2 = 300\,kN \cdot m$$

$$e_i = \frac{M}{N} + e_a = \cdots = 220\,mm > 0.3h_0 = 138\,mm$$

大偏压

$$\cdots \quad C_1 b^2 f_c (1 - \xi)$$

$$A_s' = \frac{N_e - \alpha_1 f_c h_0 \xi_b (1-\xi_b)}{f_y'(h_0 - a_s')} = \cdots = 1961.4\ mm^2$$

$\xi = \xi_b$? 考原因

$$A_{s,min}' = \frac{0.2}{100} bh = \cdots = 300\ mm^2 \Rightarrow 4\Phi 25,\ A_s' = 1963\ mm^2$$

$$A_s = \frac{\alpha_1 f_c b h_0 \xi_b + f_y' A_s' - N}{f_y} = \cdots = 634.2\ mm^2$$

$$A_{s,min} = \cdots = 300\ mm^2 \Rightarrow 2\Phi 22 \quad 760\ mm^2$$

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$$\rho = \frac{A_s + A_s'}{A} = 1.75\% > 0.55\%$$

7.4 $\frac{l_0}{b} = \frac{4500}{300} = 15 \Rightarrow \varphi = 0.895$

$$N_u = 0.9\varphi(f_c A + f_y' A_s') = \cdots 2480.5\ kN > N = 1500\ kN$$

整理数据
↓

二阶效应

↓

大小偏压

↓

计算

↓

验算（4个），3个配筋，1个承载力

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7.5 1° $\alpha_1 = 1.0$. $f_c = 19.1\,MPa$ $f_t = 1.71\,MPa$

$f_y = f_y' = 360\,MPa$.

$a_s = 40\,mm$ $h_0 = 460\,mm$

2° $\frac{M_1}{M_2}=1>0.9$ 二阶效应

$C_m=0.7+0.3\frac{M_1}{M_2}=\cdots=1$

$\zeta_c=\cdots=1.1>1$ $\zeta_0=1$

$\eta_{ns}=\cdots=\cdots=1.018 \Rightarrow C_m\eta_{ns}=1.018$

3° $x=\frac{N}{\alpha_1 f_c b}=\cdots=22.7mm<\xi_b h_0=\cdots=238.3mm$

$\Rightarrow$ 大偏压

$x<2a_s'=80mm$

4° $e_0=\cdots=1644.6mm$ $e_i=e_0+e_a=1664.6mm$

$e'=e_i-\frac{h}{2}+a_s'=\cdots=1454.6mm$

$x=80mm$ Why?

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$$\Rightarrow A_s = A_s' = \frac{Ne}{f_y(h_0 - a_s')} = \cdots = 1250.6\ mm^2$$

4 ⊥ 20 $\quad A_s = A_s' = 1257\ mm^2$

5° $\rho_{侧}$ $\quad \rho_{全截面}$ $\quad N_u$

7.6 $\alpha_1 = 1.0$ $\quad f_c = 19.1\ MPa$ $\quad f_c = 1.71\ MPa$

$f_y = f_y' = 360\ MPa$

$a_s = 40\ mm$ $\quad h_0 = \cdots = 560\ mm$

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2° $\frac{M_1}{M_2} = \cdots = 1 \Rightarrow$ 二阶效应

$$C_m = 0.7 + 0.3\frac{M_1}{M_2} = 1$$

$$\zeta_c = \frac{0.5 f_c A}{N} = \cdots = 0.573$$

$$\eta_{ns} = \cdots = \cdots = 1.35 \Rightarrow C_m \eta_{ns} = 1.35$$

$M = C_m \eta_{ns} M_2 = 270\ kN\cdot m$

3° $x = \dfrac{N}{\alpha_1 f_c b} = 523.6 > \xi_b h_0 = 290.1\ mm$

小偏压

4° $e_i = e_0 + e_a = \cdots = 87.5\ mm$

$e = \cdots = 347.5\ mm$

$$\xi = \frac{N - \xi_b \alpha_1 f_c b h_0}{\dfrac{Ne - 0.43\alpha_1 f_c b h_0^2}{(\beta_1 - \xi_b)(h_0 - a_s')} + \alpha_1 f_c b h_0} + \xi_b = \cdots = 0.783$$

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$$A_s = A_s' = \frac{Ne - \alpha_1 f_c b h_0^2 \xi (1 - 0.5\xi)}{f_y' (h_0 - a_s')} = \cdots = 1330\ mm^2$$

4Φ22 $A_s = A_s' = 1521\ mm^2$

5° 验算 ρ_{min} $\rho_{全}$ N_u

7-7 $h_0 = 560\ mm$ $e_0 = 500\ mm$

$N_u = \alpha_1 f_c b h_0 \xi$

$N_b = \alpha_1 f_c b h_0 \xi_b = \cdots = 2216.2\ kN > N$

大偏压

$$x = \frac{N}{\alpha_1 f_c b} = \cdots = 62.2\ mm < 2a_s' \Rightarrow x = 80\ mm$$

$$A_s' = A_s = \frac{N e'}{f_y (h_0 - a_s)} = \cdots = 660\ mm^2$$

$\Rightarrow 2\Phi 22$

验算

7-8 $h_0 = h - a_s = 460\ mm$ $e_a = \frac{h}{30} = 16.7 \Rightarrow e_a = 20\ mm$

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$e_i = e_0 + e_a = \cdots = 215 > 0.3 h_0 = 138$ 大偏压

$$\begin{cases} N = \alpha_1 f_c b x \\ Ne = \alpha_1 f_c b x (h_0 - \frac{1}{2} x) + f_y' A_s' (h_0 - a_s') \end{cases}$$

$x = \cdots = 202.6\ mm > 2a_s'$

$x < \xi_b h_0 = 238\ mm$

$N_u = \alpha_1 f_c b x$. Ne

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